

INTRODUCTION TO NUMBER SYSTEMS

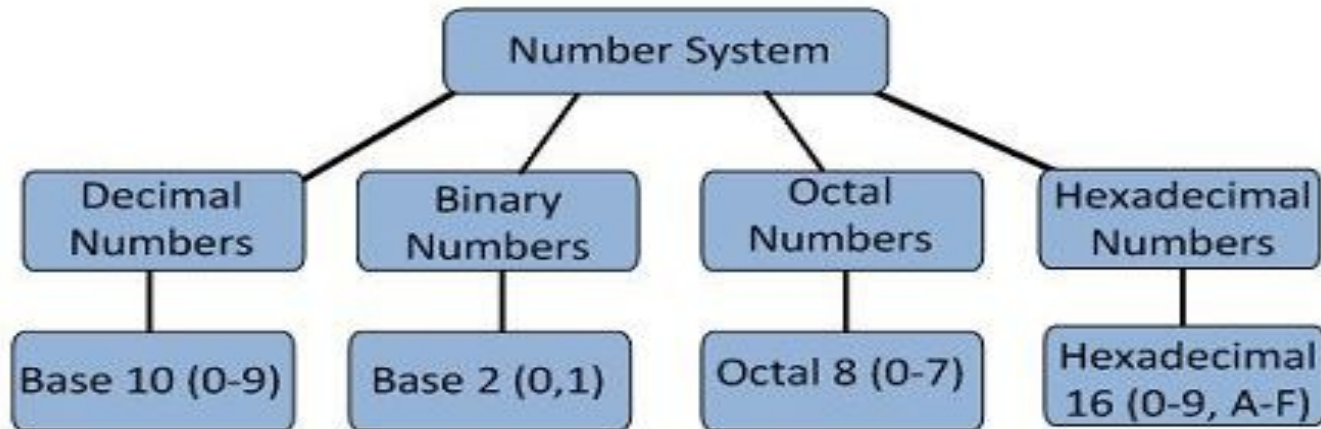
- In digital electronics, the number system is used for representing the information.
- The number system has different bases and the most common of them are the decimal, binary, octal, and hexadecimal.
- The **base or radix** of the number system is the total number of the digit used in the number system.
- Suppose if the number system representing the digit from 0 – 9 then the base of the system is the 10.

NUMBER SYSTEMS

Types of Number Systems

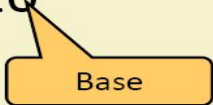
Some of the important types of number system are:

- Decimal Number System
- Binary Number System
- Octal Number System
- Hexadecimal Number System



DECIMAL NUMBER SYSTEMS

- The number system is having digit 0, 1, 2, 3, 4, 5, 6, 7, 8, 9;
- This number system is known as a decimal number system because total ten digits are involved.
- The base of the decimal number system is 10.

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$


BINARY NUMBER SYSTEMS

- The modern computers do not process decimal number; they work with another number system known as a binary number system which uses only two digits 0 and 1.
- The base of binary number system is 2 because it has only two digit 0 and 1.
- The digital electronic equipments work on the binary number system and hence the decimal number system is converted into binary system.
- The table is shown on the next slide contains:
the decimal, binary, octal, and hexadecimal representation from 0 to 15.

BINARY NUMBER SYSTEMS

Decimal	Binary	Octal	Hexa-decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

OCTAL NUMBERS

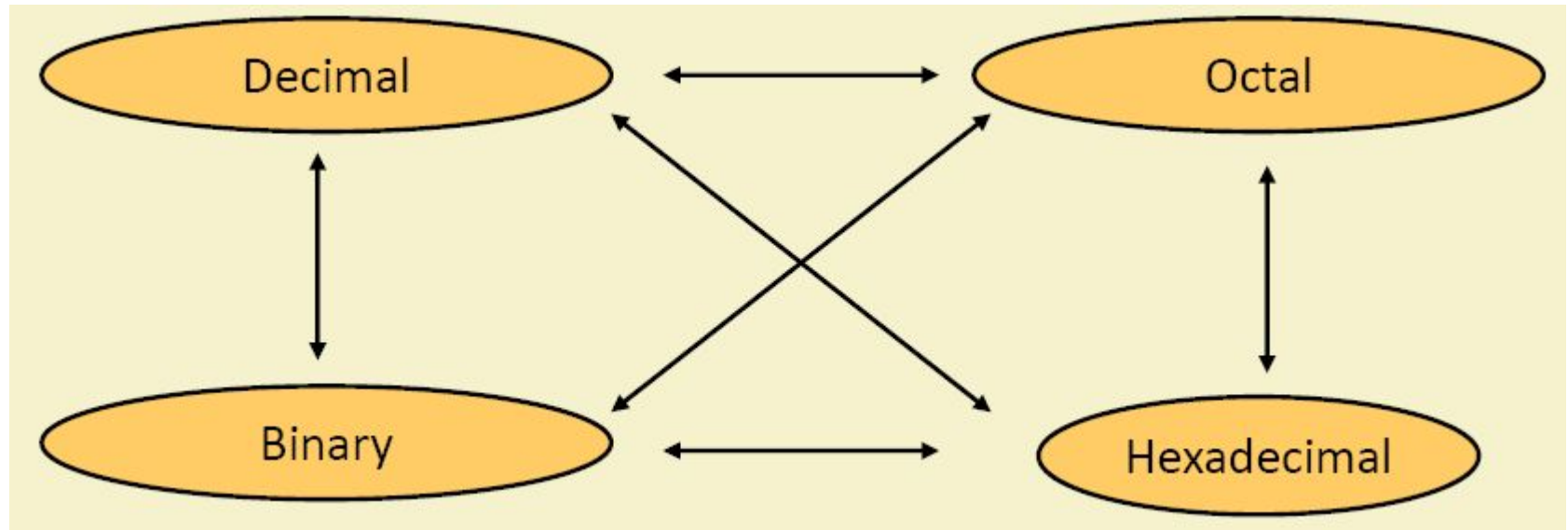
- The base of a number system is equal to the number of digits used, i.e., for decimal number system the base is ten while for the binary system the base is two. The octal system has the base of eight as it uses eight digits 0, 1, 2, 3, 4, 5, 6, 7.
- All these digits from 0 to 7 have the same physical meaning as by decimal symbols, the next digit in the octal number is represented by 10, 11, 12... which are equivalent to decimal digits 8, 9, 10... respectively.

HEXADECIMAL NUMBERS

- These numbers are used extensively in microprocessor. The hexadecimal number system has a base of 16, and hence it consists of the following sixteen number of digits.
- 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
- The size of the hexadecimal is much shorter than the binary number which makes them easy to write and remember.
- Let 00 to 0F representing hexadecimal numbers from 0 to 15, then 10, 11, 12... in hex will represent 16, 17, 18...in decimal.

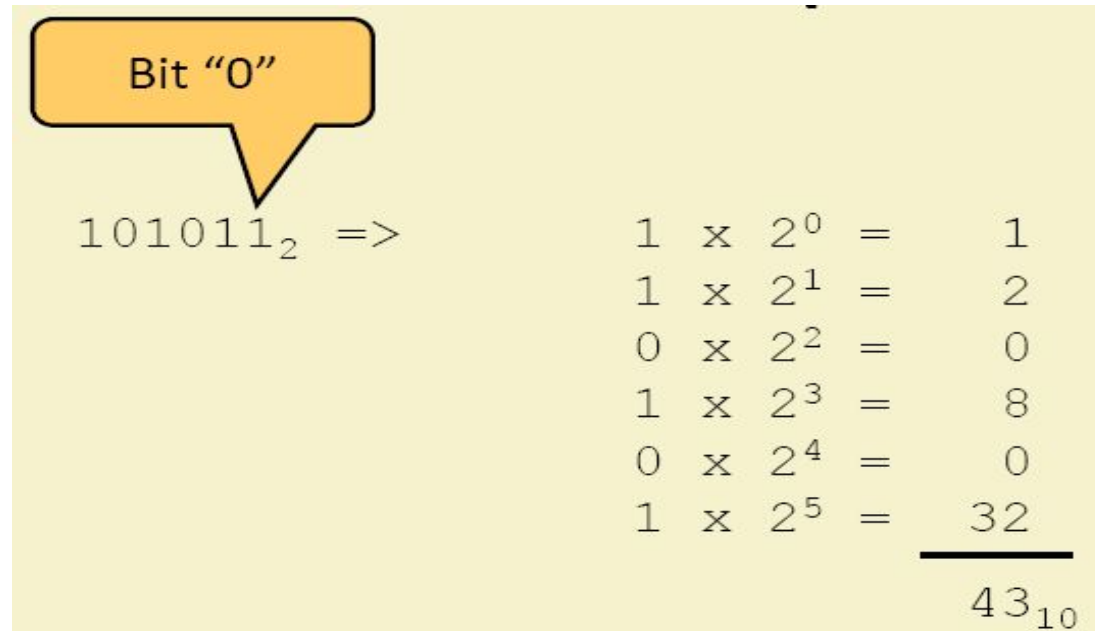
CODE CONVERSION

Conversion among base:



BINARY TO DECIMAL CONVERSION:

- Multiply each bit by 2^n , where n is “weight” of the bits
- The weight is the position of the bit starting from 0, moving from right to left.
- Add the result
- Example:



Bit "0"

$101011_2 \Rightarrow$

1	x	2^0	=	1
1	x	2^1	=	2
0	x	2^2	=	0
1	x	2^3	=	8
0	x	2^4	=	0
1	x	2^5	=	32
				43_{10}

OCTAL TO DECIMAL

- multiplying each digit by 8^n bits where n the weight of the digits.
- the weight is the position of the digit starting from 0. moving from right to left.
- add the result
- example:

$724_8 \Rightarrow$

$$\begin{array}{rcl} 4 & \times & 8^0 = & 4 \\ 2 & \times & 8^1 = & 16 \\ 7 & \times & 8^2 = & 448 \\ \hline & & & 468_{10} \end{array}$$

HEXADECIMAL TO DECIMAL CONVERSION

- Multiplying each digit by 16^n bits where n the weight of the digits.
- The weight is the position of the digit starting from 0, moving from right to left.
- Add the result
- Example:

$$\begin{array}{rcllcll} ABC_{16} => & C & \times & 16^0 & = & 12 & \times & 1 & = & 12 \\ & B & \times & 16^1 & = & 11 & \times & 16 & = & 176 \\ & A & \times & 16^2 & = & 10 & \times & 256 & = & 2560 \\ & & & & & & & & & \hline & & & & & & & & & 2748_{10} \end{array}$$

BINARY TO DECIMAL CONVERSION

Example : Convert $(10101)_2$ to decimal.

$$\begin{aligned}10101 &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\&= 16 + 0 + 4 + 0 + 1 \\&= 21\end{aligned}$$

$$(10101)_2 = (21)_{10}$$

Example : Convert $(11011.101)_2$ to decimal.

$$\begin{aligned}11011.101 &= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) + (1 \times 2^{-1}) + (0 \times 2^{-2}) + \\&\quad (1 \times 2^{-3}) \\&= 16 + 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125 \\&= 27.625\end{aligned}$$

$$(11011.101)_2 = (27.625)_{10}$$

DECIMAL TO BINARY CONVERSION

Example : convert $(52)_{10}$ to binary.

Sol:

2	52
2	26
2	13
2	6
2	3
2	1
	0

Remainder

0

0

1

0

1

1



$$(52)_{10} = (110100)_2$$

DECIMAL TO BINARY CONVERSION

Decimal Fraction Number to Binary

Example 1.3

Convert $(0.6875)_{10}$ to binary. The multiplied-by-2 process is continued until the fraction becomes 0 or until the number of digits has sufficient accuracy.

	Integer		Fraction		Coefficient
$0.6875 \times 2 =$	1	+	0.3750	$a_{-1} =$	1
$0.3750 \times 2 =$	0	+	0.7500	$a_{-2} =$	0
$0.7500 \times 2 =$	1	+	0.5000	$a_{-3} =$	1
$0.5000 \times 2 =$	1	+	0.0000	$a_{-4} =$	1



$$(0.6875)_{10} = (0.a_{-1}a_{-2}a_{-3}a_{-4})_2 = (0.1011)_2$$

- ♣ To convert a decimal fraction to a base- r number, a similar procedure is used. However, multiplication is by r instead of 2, and the coefficients found from the integers may range in value from 0 to $r - 1$ instead of 0 and 1.

DECIMAL TO OCTAL CONVERSION

Example : Convert $(378.93)_{10}$ to octal

8	378
8	47
8	5
	0

Remainder

2

7

5



$$0.93 \times 8 = 7.44$$

$$0.44 \times 8 = 3.52$$

$$0.52 \times 8 = 4.16$$

$$0.16 \times 8 = 1.28$$

So, $0.93_{10} = 0.7341_8$

7

3

4

1



Ans: $(378.93)_{10} = (572.7341)_8$

DECIMAL TO HEXADECIMAL CONVERSION

Example : Convert $(59575)_{10}$ to hex

16	59575	7
16	1590	11 $\Rightarrow$ B
16	232	8
16	14	14 $\Rightarrow$ E



$$(59575)_{10} = (\text{E8B7})_{16}$$

BINARY TO OCTAL CONVERSION

Binary Number: 101010011.110100

Group of three digits: $\begin{array}{ccccc} \underline{101} & \underline{010} & \underline{011} & \underline{110} & \underline{100} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{array}$

Octal Equivalent: 5 2 3 6 4

= (523.64)₈

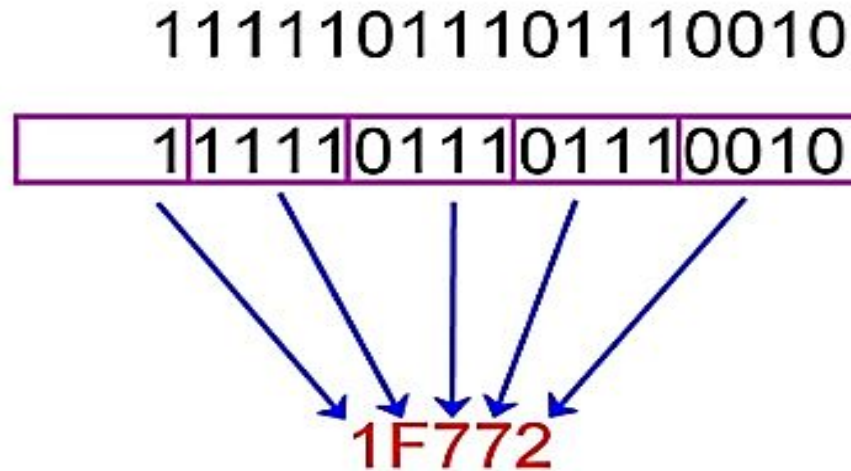
OCTAL TO BINARY CONVERSION

$$\begin{array}{ccc} \mathbf{1} & \mathbf{2} & \mathbf{.5} \\ \swarrow & \searrow & \swarrow \\ \mathbf{001} & \mathbf{010} & \mathbf{101} \end{array}$$
$$(12.5)_8 = (001010.101)_2$$

$$\begin{array}{ccc} \mathbf{7} & \mathbf{2} & \mathbf{1} \\ \swarrow & \searrow & \swarrow \\ \mathbf{111} & \mathbf{010} & \mathbf{001} \end{array}$$
$$(721)_8 = (111010001)_2$$

BINARY TO HEXADECIMAL CONVERSION

Converting Binary to Hex



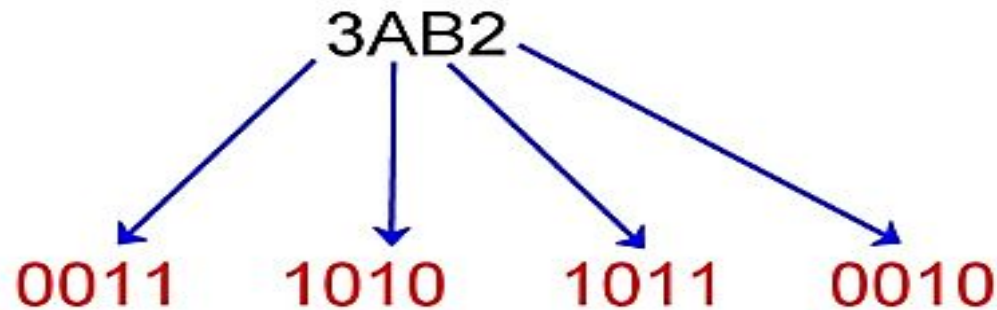
Starting from the right, divide up into blocks of 4 digits

Write down the hex value of each block

$11111011101110010_2 = 1F772_{16}$

HEXADECIMAL TO BINARY CONVERSION

Converting Hex to Binary



$$3AB2_{16} = 11101010110010_2$$

Old Questions

- Determine the value of x if $(193)_x = (623)_8$
- Convert $(100000011110)_2$ into hexadecimal and octal number system without converting it to decimal.
- The solution to the quadratic equation $k^2 - 11k + 22 = 0$ are $k = 3$ and $k = 6$. What are the base of number systems?
- Convert $(238.99)_{10}$ into binary, hexadecimal and octal.

Recap

- The number system has different bases and the most common of them are the decimal, binary, octal, and hexadecimal.
- The **base or radix** of the number system is the total number of the digit used in the number system.
- Conversion of the important types of number system from one another:

Decimal Number System

Binary Number System

Octal Number System

Hexadecimal Number System

Binary Arithmetic

- Addition:

$0 + 0 = 0$	Sum of 0 with a carry of 0
$0 + 1 = 1$	Sum of 1 with a carry of 0
$1 + 0 = 1$	Sum of 1 with a carry of 0
$1 + 1 = 10$	Sum of 0 with a carry of 1

(a)	$\begin{array}{r} 11 \\ + 11 \\ \hline 110 \end{array}$	$\begin{array}{r} 3 \\ + 3 \\ \hline 6 \end{array}$	(b)	$\begin{array}{r} 100 \\ + 10 \\ \hline 110 \end{array}$	$\begin{array}{r} 4 \\ + 2 \\ \hline 6 \end{array}$
(c)	$\begin{array}{r} 111 \\ + 11 \\ \hline 1010 \end{array}$	$\begin{array}{r} 7 \\ + 3 \\ \hline 10 \end{array}$	(d)	$\begin{array}{r} 110 \\ + 100 \\ \hline 1010 \end{array}$	$\begin{array}{r} 6 \\ + 4 \\ \hline 10 \end{array}$

Binary Arithmetic

- Subtraction:

$$0 - 0 = 0$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$10 - 1 = 1 \quad 0 - 1 \text{ with a borrow of } 1$$

Perform the following binary subtractions:

(a) $11 - 01$

(b) $11 - 10$

Binary Arithmetic

- Example:

Perform the following binary subtractions:

(a) $11 - 01$ (b) $11 - 10$

Sol:

(a)	$\begin{array}{r} 11 \\ -01 \\ \hline 10 \end{array}$	$\begin{array}{r} 3 \\ -1 \\ \hline 2 \end{array}$	(b)	$\begin{array}{r} 11 \\ -10 \\ \hline 01 \end{array}$	$\begin{array}{r} 3 \\ -2 \\ \hline 1 \end{array}$
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Binary Arithmetic

- Multiplication:

$$\begin{array}{l} 0 \times 0 = 0 \\ 0 \times 1 = 0 \\ 1 \times 0 = 0 \\ 1 \times 1 = 1 \end{array}$$

Example: (a) 11×11 (b) 101×111

Solution

(a)

$$\begin{array}{r} 11 \qquad 3 \\ \times 11 \quad \times 3 \\ \hline 11 \qquad 9 \\ +11 \qquad \\ \hline 1001 \end{array}$$

Partial products {

(b)

$$\begin{array}{r} 111 \qquad 7 \\ \times 101 \quad \times 5 \\ \hline 111 \qquad 35 \\ 000 \\ +111 \\ \hline 100011 \end{array}$$

Partial products {

Binary Arithmetic

- **Division:** follow same procedure as decimal number system

Example:

Perform the following binary divisions:

(a) $110 \div 11$

$$\begin{array}{r} 10 \\ 11 \overline{)110} \\ \underline{11} \\ 000 \end{array}$$

(b) $110 \div 10$

$$\begin{array}{r} 11 \\ 10 \overline{)110} \\ \underline{10} \\ 10 \\ \underline{10} \\ 00 \end{array}$$

COMPLEMENT

Complements of Numbers

- ♣ Complements are used in digital systems to simplify the subtraction operation
- ♣ There are two types of complements for each base- r system: the radix complement and diminished radix complement.

➡ the r 's complement and the second as the $(r - 1)$'s complement.

■ Diminished Radix Complement

Given a number N in base r having n digits, the $(r - 1)$'s complement of N is defined as $(r^n - 1) - N$. For decimal numbers, $r = 10$ and $r - 1 = 9$, so the 9's complement of N is $(10^n - 1) - N$.

Example:

The 9's complement of 546700 is $999999 - 546700 = 453299$.

The 9's complement of 012398 is $999999 - 012398 = 987601$.

- ♣ For binary numbers, $r = 2$ and $r - 1 = 1$, so the 1's complement of N is $(2^n - 1) - N$.

Example:

★ The 1's complement of 1011000 is 0100111

The 1's complement of 0101101 is 1010010

COMPLEMENT

■ Radix Complement

The r 's complement of an n -digit number N in base r is defined as $r^n - N$ for $N \neq 0$ and as 0 for $N = 0$. Comparing with the $(r - 1)$'s complement, we note that the r 's complement is obtained by adding 1 to the $(r - 1)$'s complement, since $r^n - N = [(r^n - 1) - N] + 1$.

Example: Base-10

The 10's complement of 012398 is 987602
The 10's complement of 246700 is 753300

Example: Base-2

The 2's complement of 1101100 is 0010100
The 2's complement of 0110111 is 1001001



SIGNED BINARY NUMBERS

Table 1.3
Signed Binary Numbers

Decimal	Signed-2's Complement	Signed-1's Complement	Signed Magnitude
+7	0111	0111	0111
+6	0110	0110	0110
+5	0101	0101	0101
+4	0100	0100	0100
+3	0011	0011	0011
+2	0010	0010	0010
+1	0001	0001	0001
+0	0000	0000	0000
-0	—	1111	1000
-1	1111	1110	1001
-2	1110	1101	1010
-3	1101	1100	1011
-4	1100	1011	1100
-5	1011	1010	1101
-6	1010	1001	1110
-7	1001	1000	1111
-8	1000	—	—
$-(2^{(n-1)}) \text{ to } (2^{(n-1)}-1)$ $-(2^{(n-1)}-1) \text{ to } (2^{(n-1)}-1)$ $-(2^{(n-1)}-1) \text{ to } (2^{(n-1)}-1)$			

SUBTRACTION WITH COMPLEMENT

Subtraction with Complements

- The subtraction of two n -digit unsigned numbers $M - N$ in base r can be done as follows:
 - Add the minuend M to the r 's complement of the subtrahend N .
 $\rightarrow M + (r^n - N) = M - N + r^n$
 - $M \geq N \rightarrow$ the sum will produce an end carry r^n , which can be discarded, what is left is the result $M - N$
 - $M < N \rightarrow$ the sum does not produce an end carry, and is equal to $r^n - (N - M)$

SUBTRACTION WITH COMPLEMENT

10's Complement Subtraction

Example 1.5

Using 10's complement, subtract $72532 - 3250$.

	$M =$	72532
10's complement of	$N =$	<u>+96750</u>
	Sum =	169282
	Discard end carry $10^5 =$	<u>-100000</u>
	Answer =	69282

Example 1.6

Using 10's complement, subtract $3250 - 72532$

	$M =$	03250
10's complement of	$N =$	<u>+27468</u>
	Sum =	30718



There is no end carry.



Therefore, the answer is $-(10's \text{ complement of } 30718) = -69282$.

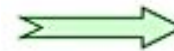
SUBTRACTION WITH COMPLEMENT

2's Complement Subtraction

Example 1.7

Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$ and (b) $Y - X$ by using 2's complement.

(a)	$X =$	1010100
	2's complement of $Y =$	<u>+0111101</u>
	Sum =	10010001
	Discard end carry $2^7 =$	<u>-10000000</u>
	Answer. $X - Y =$	0010001



The end carry says that the answer is positive.

(b)	$Y =$	1000011
	2's complement of $X =$	<u>+0101100</u>
	Sum =	1101111



There is no end carry. Therefore, the answer is $Y - X = -$ (2's complement of 1101111) = - 0010001.

SUBTRACTION WITH COMPLEMENT

1's Complement Subtraction

- ♣ Subtraction of unsigned numbers can also be done by means of the $(r - 1)$'s complement. Remember that the $(r - 1)$'s complement is one less than the r 's complement.

Example 1.8

Repeat Example 1.7, but this time using 1's complement.

(a) $X - Y = 1010100 - 1000011$	
$X =$	1010100
1's complement of $Y =$	± 0111100
Sum =	10010000
End-around carry =	$\begin{array}{r} + \quad \quad 1 \\ \hline \end{array}$
Answer. $X - Y =$	0010001

The end carry
need to be added
in the final result.

(b) $Y - X = 1000011 - 1010100$	
$Y =$	1000011
1's complement of $X =$	$+ 0101011$
Sum =	1101110

There is no end carry,
Therefore, the answer is
 $Y - X = - (1's \text{ complement of } 1101110) = - 0010001.$

SIGNED BINARY NUMBERS

Signed Binary Numbers

- ♣ To represent negative integers, we need a notation for negative values.
- ♣ It is customary to represent the sign with a bit placed in the leftmost position of the number.
- ♣ The convention is to make the sign bit 0 for positive and 1 for negative.

Example: Although there is only one way to represent +9, there are three different ways to represent -9 with eight bits:

Signed-magnitude representation:	10001001
Signed-1's-complement representation:	11110110
Signed-2's-complement representation:	11110111



- ♣ **Table 3** lists all possible four-bit signed binary numbers in the three representations.

Boolean algebra & Logic Gates

- Boolean Algebra is used to analyze and simplify the digital (logic) circuits. It uses only the binary numbers i.e. 0 and 1. It is also called as **Binary Algebra** or **logical Algebra**. Boolean algebra was invented by **George Boole** in 1854.
- Binary logic consists of binary variables and a set of logical operations. The variables are designated by letters of the alphabet, such as A, B, C, x, y, z, etc, with each variable having two and only two distinct possible values: 1 and 0, There are three basic logical operations: AND, OR, and NOT.
- **Variable:** A symbol used to represent an action/data
- **Complement:** Inverse of the Variable
- **Literal:** A variable or complement of a variable

Boolean algebra & Logic Gates

Following are the important rules used in Boolean algebra.

- Variable used can have only two values. Binary 1 for HIGH and Binary 0 for LOW.
- Complement of a variable is represented by an overbar (-).
- ORing of the variables is represented by a plus (+) sign between them. For example ORing of A, B, C is represented as $A + B + C$.
- Logical ANDing of the two or more variable is represented by writing a dot between them such as $A.B.C$. Sometime the dot may be omitted like ABC .

Boolean algebra & Logic Gates

There are six types of Boolean Laws.

1) Commutative law:

- Any binary operation which satisfies the following expression is referred to as commutative operation.

$$(i) A.B = B.A$$

$$(ii) A + B = B + A$$

Commutative law states that changing the sequence of the variables does not have any effect on the output of a logic circuit.

2) Associative law:

This law states that the order in which the logic operations are performed is irrelevant as their effect is the same.

$$(i) (A.B).C = A.(B.C)$$

$$(ii) (A + B) + C = A + (B + C)$$

Boolean algebra & Logic Gates

- **Distributive law:**

Distributive law states the following condition.

$$A.(B + C) = A.B + A.C$$

AND law

- These laws use the AND operation. Therefore they are called as **AND** laws.

$$(i) A.0 = 0$$

$$(ii) A.1 = A$$

$$(iii) A.A = A$$

$$(iv) A.\overline{A} = 0$$

OR law

- These laws use the OR operation. Therefore they are called as **OR** laws

$$(i) A + 0 = A$$

$$(ii) A + 1 = 1$$

$$(iii) A + A = A$$

$$(iv) A + \overline{A} = 1$$

Boolean algebra & Logic Gates

The Duality Principle

Steps/Rules:

Given any true Boolean expression,

Step1: *swapping $AND \leftrightarrow OR$*

Step2: *$0 \leftrightarrow 1$ — while*

Step3: *leaving variables untouched.*

Produces another true expression, called its dual.

Boolean algebra & Logic Gates

Original	Dual
$A \cdot 1 = A$	$A + 0 = A$
$A \cdot 0 = 0$	$A + 1 = 1$
$A + A = A$	$A \cdot A = A$
$A + A' = 1$	$A \cdot A' = 0$
$A(B + C) = AB + AC$	$A + BC = (A+B)(A+C)$

De Morgan's Theorem

Statement: The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables and vice-versa.

- Complements the whole expression
- Swap AND $\leftrightarrow$ OR
- Complements every individual variable

Boolean Theorems

Table 2-1 from book Morris Mano ed.3

Postulates and Theorems of Boolean Algebra

Postulate 2	(a) $x + 0 = x$	(b) $x \cdot 1 = x$
Postulate 5	(a) $x + x' = 1$	(b) $x \cdot x' = 0$
Theorem 1	(a) $x + x = x$	(b) $x \cdot x = x$
Theorem 2	(a) $x + 1 = 1$	(b) $x \cdot 0 = 0$
Theorem 3, involution	$(x')' = x$	
Postulate 3, commutative	(a) $x + y = y + x$	(b) $xy = yx$
Theorem 4, associative	(a) $x + (y + z) = (x + y) + z$	(b) $x(yz) = (xy)z$
Postulate 4, distributive	(a) $x(y + z) = xy + xz$	(b) $x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a) $(x + y)' = x'y'$	(b) $(xy)' = x' + y'$
Theorem 6, absorption	(a) $x + xy = x$	(b) $x(x + y) = x$

Boolean algebra & Logic Gates

- **AND Gate**
- A circuit which performs an AND operation is shown in figure. It has n input ($n \geq 2$) and one output.
- **Logic diagram**



- **Truth Table**

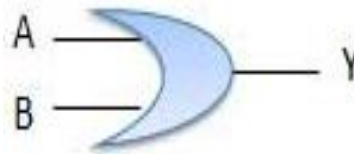
Inputs		Output
A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

Boolean algebra & Logic Gates

OR Gate

- A circuit which performs an OR operation is shown in figure. It has n input ($n \geq 2$) and one output.

Logic diagram



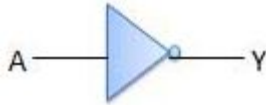
Truth Table

Inputs		Output
A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

Boolean algebra & Logic Gates

- **NOT Gate**
- NOT gate is also known as **Inverter**. It has one input A and one output Y.

- Symbol



- Truth Table

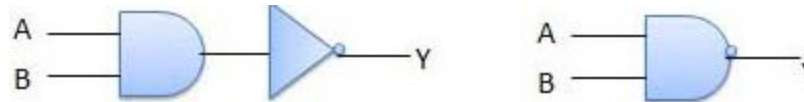
Inputs	Output
A	B
0	1
1	0

Boolean algebra & Logic Gates

NAND Gate

A NOT-AND operation is known as NAND operation. It has n input ($n \geq 2$) and one output.

- **Logic diagram**

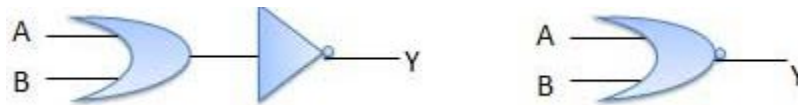


- **Truth Table**

Inputs		Output
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

Boolean algebra & Logic Gates

- **NOR Gate**
- A NOT-OR operation is known as NOR operation. It has n input ($n \geq 2$) and one output.
- Logic diagram



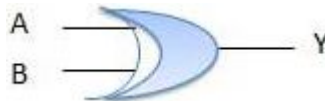
Truth Table

Inputs		Output
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Boolean algebra & Logic Gates

- **XOR Gate:**
- XOR or Ex-OR gate is a special type of gate. It can be used in the half adder, full adder and subtractor. The exclusive-OR gate is abbreviated as EX-OR gate or sometime as X-OR gate. It has n input ($n \geq 2$) and one output.

- **Logic diagram**



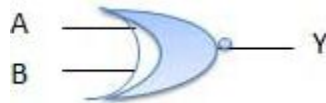
- **Truth Table**

Inputs		Output
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Boolean algebra & Logic Gates

- **XNOR Gate**
- XNOR gate is a special type of gate. It can be used in the half adder, full adder and subtractor. The exclusive-NOR gate is abbreviated as EX-NOR gate or sometime as X-NOR gate. It has n input ($n \geq 2$) and one output.

- **Logic diagram**



- **Truth Table**

Inputs		Output
A	B	$A \oplus B$
0	0	1
0	1	0
1	0	0
1	1	1

Daily Quiz

Q1. The NOR gate output will be high if the two inputs are _____

- a) **00**
- b) 01
- c) 10
- d) 11

Q2. A universal logic gate is one which can be used to generate any logic function. Which of the following is a universal logic gate?

- a) OR
- b) AND
- c) XOR
- d) **NAND**

Q3. Which of following are known as universal gates?

- a) **NAND & NOR**
- b) AND & OR
- c) XOR & OR
- d) EX-NOR & XOR

Weekly Assignment

- What do you understand by universal gates?
- Realize a Ex-or gate with the help of NAND gate.
- Realize a OR gate using NAND gates.
- How an Ex-or gate can work as an inverter?
- What do understand by the term SOP and POS?

Recap

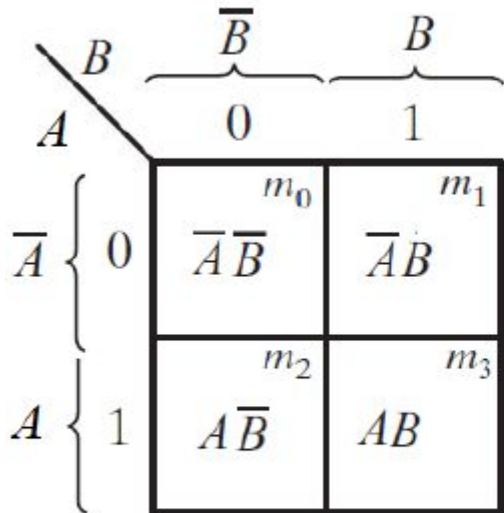
- Boolean Algebra is used to analyze and simplify the digital (logic) circuits.
- It uses only the binary numbers i.e. 0 and 1.
- It is also called as **Binary Algebra** or **logical Algebra**.
- Binary logic consists of binary variables and a set of logical operations.
- Here variable having two and only two distinct possible values: 1 and 0.
- There are three basic logical operations: AND, OR, and NOT.

KARNAUGHMAP (K-MAP) REPRESENTATION

- The complexity of the digital logic gates that implement a Boolean function is directly related to the complexity of the algebraic expression from which the function is implemented.
- The map method provides a simple straightforward procedure for minimizing Boolean functions.
- The map method, first proposed by Veitch and modified by Karnaugh, is also known as the "Veitch diagram" or the "Karnaugh map."
- The map is a diagram made up of squares. Each square represents one minterm.
- Since any Boolean function can be expressed as a sum of minterms, it follows that a Boolean function is recognized graphically in the map from the area enclosed by those squares whose min terms are included in the function.

KARNAUGHMAP (K-MAP) REPRESENTATION

- Two-Variable Karnaugh Map:



$$Y = F(A,B) = \sum m(2,3)$$

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

	$\bar{B}$	B
$\bar{A}$	0	0
A	1	1

$$Y = A$$

$$Y = F(A,B) = \sum m(1,2,3)$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

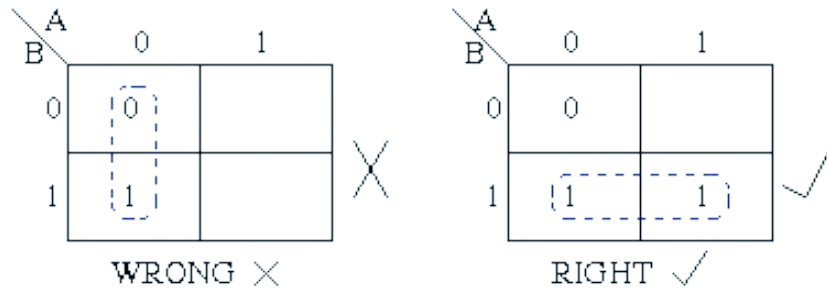
	$\bar{B}$	B
$\bar{A}$	0	1
A	1	1

$$Y = A + B$$

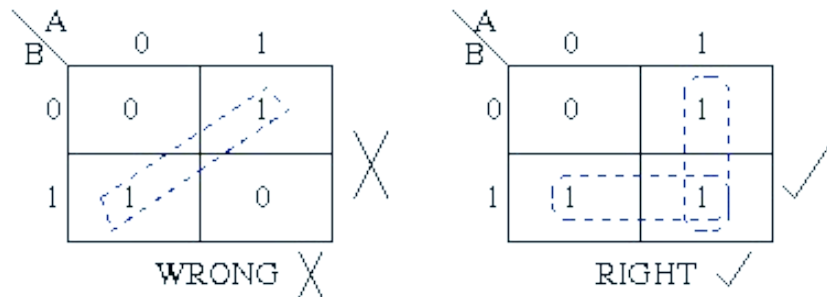
KMAP

KMAP Simplification Rules:

- Groups may not include any cell containing a zero.

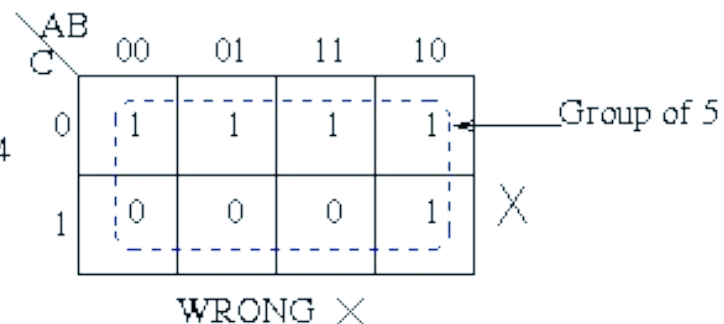
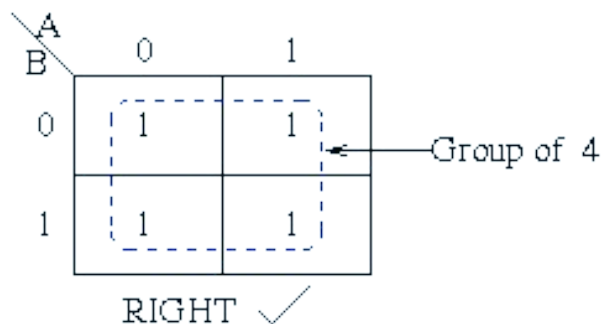
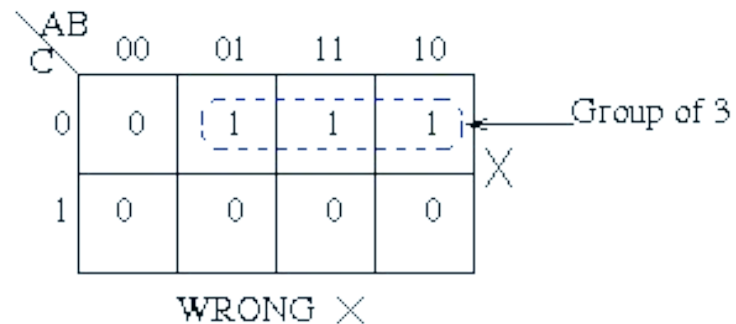
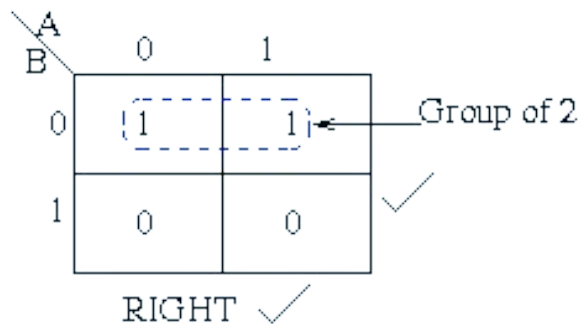


- Groups may be horizontal or vertical, but not diagonal.



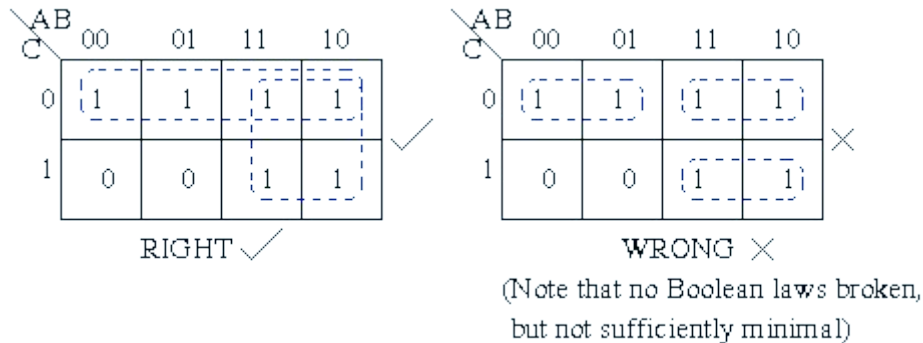
KMAP

- Groups must contain 1, 2, 4, 8, or in general 2^n cells. That is if $n = 1$, a group will contain two 1's since $2^1 = 2$.
- If $n = 2$, a group will contain four 1's since $2^2 = 4$.

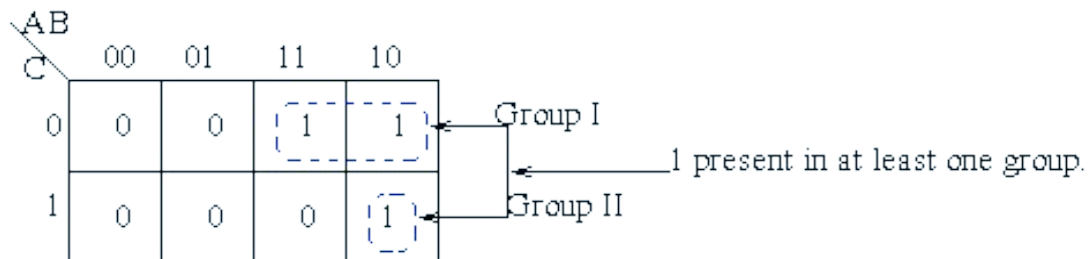


KMAP

- Each group should be as large as possible.

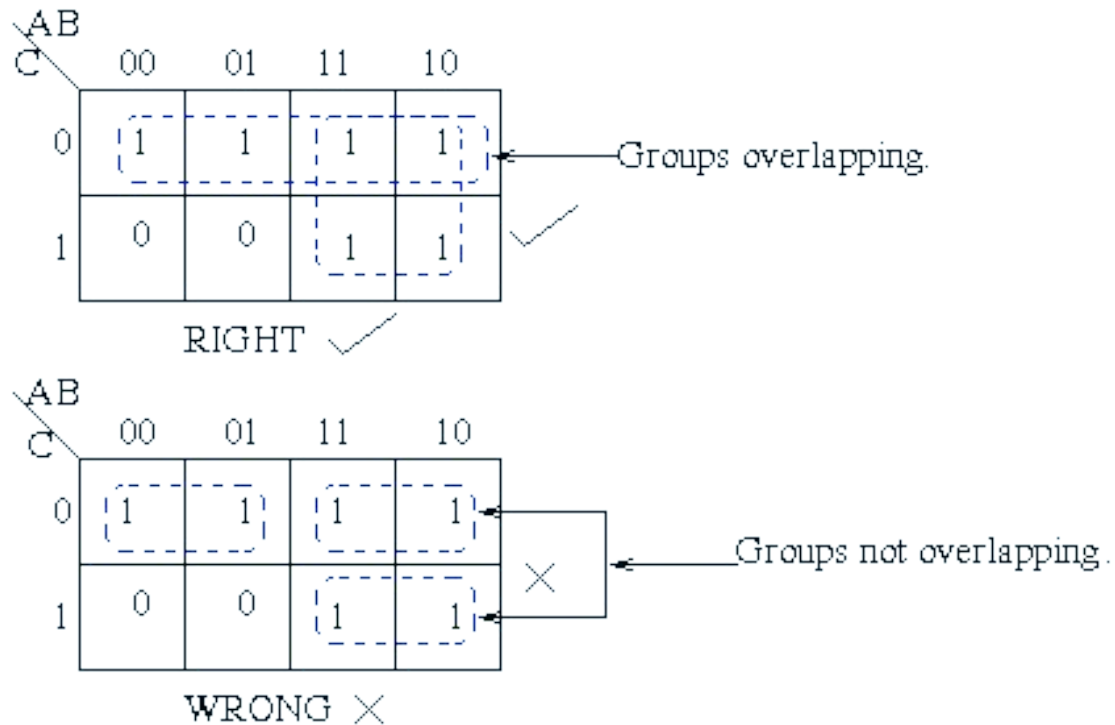


- Each cell containing a *one* must be in at least one group.



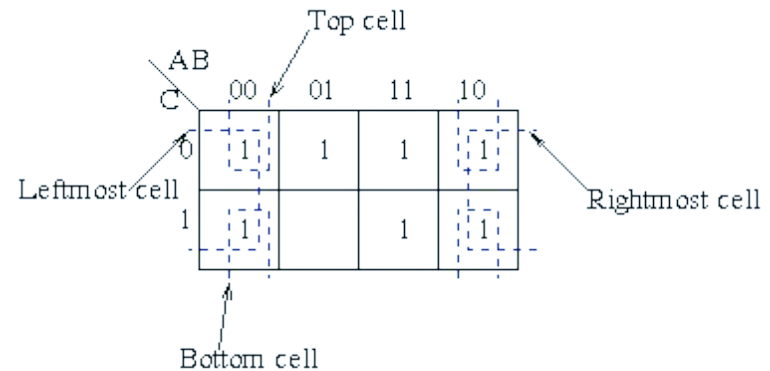
KMAP

- Groups may overlap.

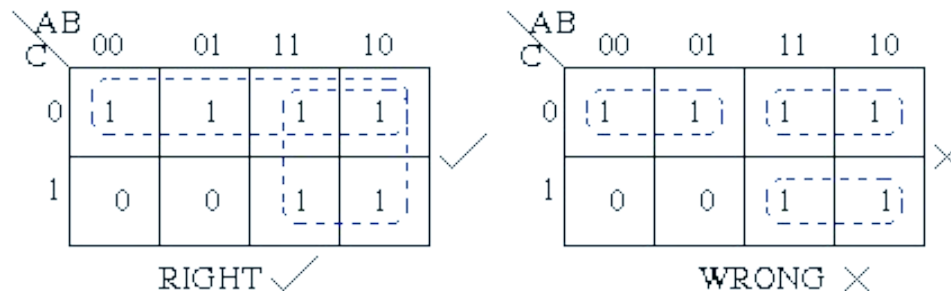


KMAP

- Groups may wrap around the table. The leftmost cell in a row may be grouped with the rightmost cell and the top cell in a column may be grouped with the bottom cell.



- There should be as few groups as possible, as long as this does not contradict any of the previous rules.



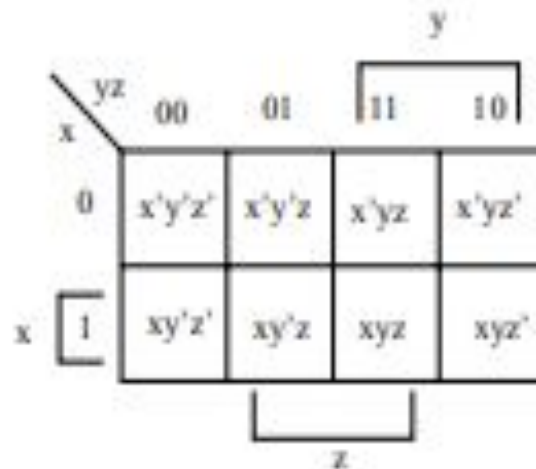
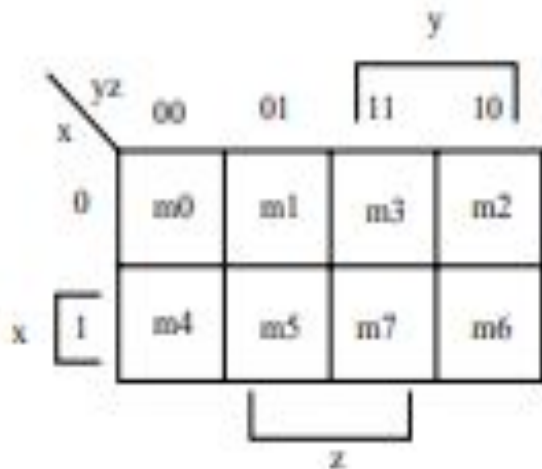
Summary:

1. No zeros allowed (In case of minterms).
2. No diagonals.
3. Only power of 2 number of cells in each group.
4. Groups should be as large as possible.
5. Every one must be in at least one group.
6. Overlapping allowed.
7. Wrap around allowed.
8. Fewest number of groups possible.

THREE VARIABLE KARNAUGH MAP

- Three-Variable Karnaugh Map:**

3 variables, $F = f(x, y, z)$



x	y	z	$F(x,y,z)$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

THREE VARIABLE KARNAUGH MAP

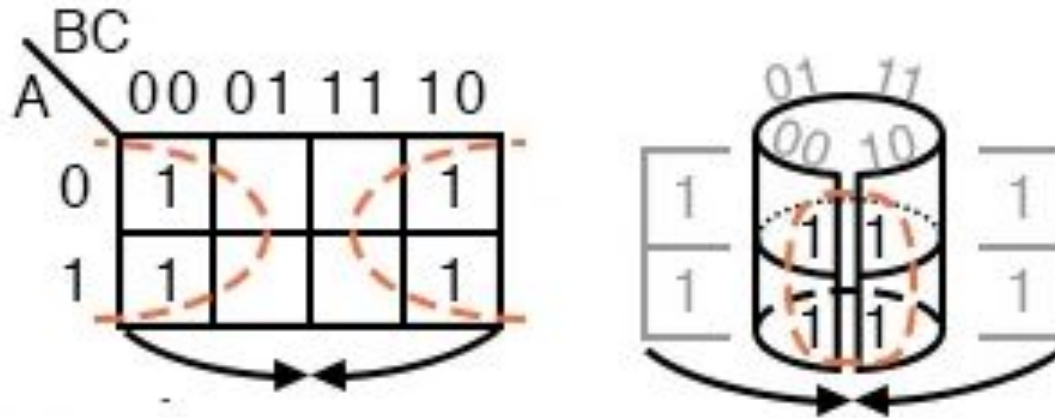
Eg: $F(A,B,C) = \sum(1,2,3,6)$ minimize the given using K-MAP.

BC A		BC			
		00	01	11	10
0	1		1	1	1
					1

Ans: $F = A'C + BC'$

THREE VARIABLE KARNAUGH MAP

Eg: $F = A'B'C' + AB'C' + A'BC' + ABC'$ minimize the given using K-MAP.



Ans: $F = C'$

THREE VARIABLE KARNAUGH MAP

Eg: $F(x, y, z) = \sum(0, 2, 4, 5, 6)$ minimize the given using K-MAP.

		y			
		z		11	10
x	0	1			1
	1	1	1		1

Ans: $F = z' + x.y'$

THREE VARIABLE KARNAUGH MAP

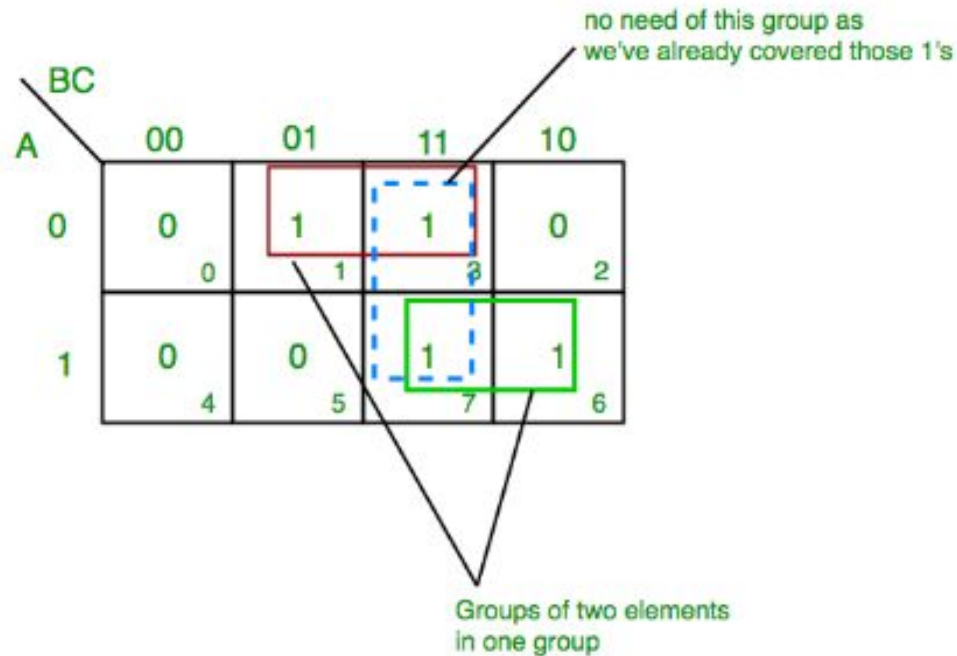
Eg: $F = \sum A, B, C(3, 5, 6, 7)$ minimize the given using K-MAP.

		BC			
		00	01	11	10
A	0			1	
	1		1	1	1

Ans: $F = BC + AC + AB$

THREE VARIABLE KARNAUGH MAP

Eg: $F(A,B,C) = \sum(1,3,6,7)$ minimize the given using K-MAP.



Ans: $F = A'C + AB$

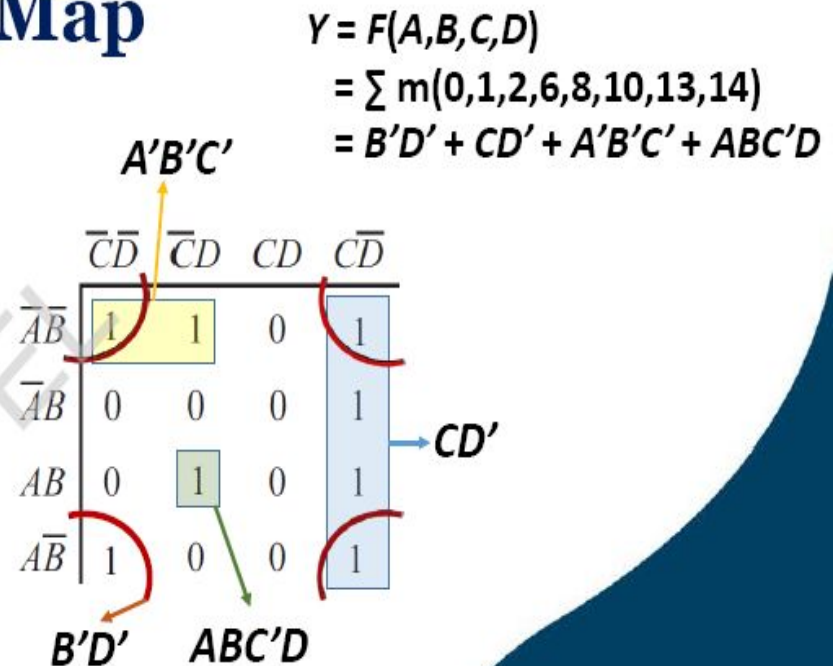
FOUR-VARIABLE KARNAUGH MAP

Four-Variable Karnaugh Map

		$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
		00	01	11	10
$\overline{A}\overline{B}$	00	0	1	3	2
$\overline{A}B$	01	4	5	7	6
AB	11	12	13	15	14
$A\overline{B}$	10	8	9	11	10

$F(A,B,C,D)$
minterm numbers

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

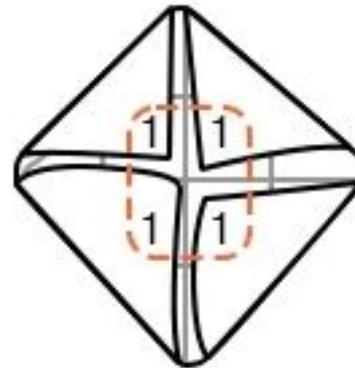


FOUR-VARIABLE KARNAUGH MAP

Eg: $F = \sum m(0, 2, 8, 10)$ minimize the given using K-MAP in SOP form.

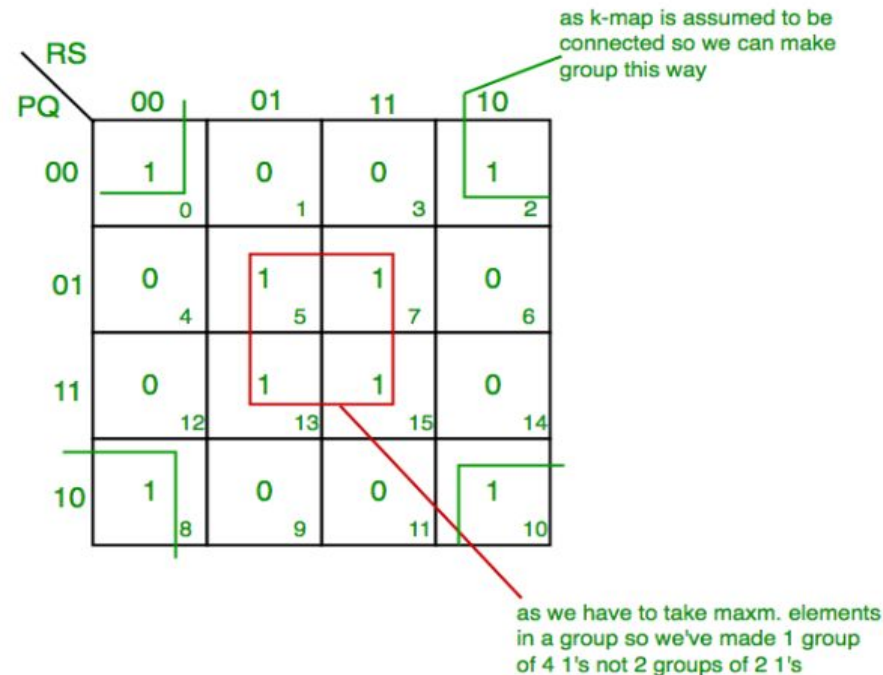
		CD			
		00	01	11	10
A B	00	1			1
	01				
	11				
	10	1			1

$$\text{Out} = \overline{B}\overline{D}$$



FOUR-VARIABLE KARNAUGH MAP

- $F(P,Q,R,S) = \sum(0,2,5,7,8,10,13,15)$ minimize the given using K-MAP.



$$F = (QS + Q'S')$$

FOUR-VARIABLE KARNAUGH MAP

- $F(A,B,C,D)=\sum(1,2,3,5,6,7,8,10,13,15)$ minimize the given using K-MAP.

$$F = \overline{A}D + \overline{A}C + BD + A\overline{B}\overline{D}$$

DON'T CARE IN KARNAUGH MAP

Design: Y is H when BCD (Binary Coded Decimal) input is odd.



$$Y = F(A, B, C, D) \\ = \sum m(1, 3, 5, 7, 9) + \\ d(10, 11, 12, 13, 14, 15)$$

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	x
1	0	1	1	x
1	1	0	0	x
1	1	0	1	x
1	1	1	0	x
1	1	1	1	x

Not considering X

	$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\overline{B}$	0	1	1	0
$\overline{A}B$	0	1	1	0
AB	x	x	x	x
$A\overline{B}$	0	1	x	x

$$Y = A'D + B'C'D$$

- If not considered, X = 0.
- If considered, X = 1.
- Consideration wherever helps.

Considering X

	$\overline{C}\overline{D}$	$\overline{C}D$	CD	$C\overline{D}$
$\overline{A}\overline{B}$	0	1	1	0
$\overline{A}B$	0	1	1	0
AB	x	x	x	x
$A\overline{B}$	0	1	x	x

$$Y = D$$

DON'T CARE IN KARNAUGH MAP

EXAMPLES OF DON'T CARE CONDITIONS (1/2)

$$F = \sum m(1, 3, 7) + \sum d(0, 5)$$

Circle the x's that help get bigger groups of 1's (or 0's if POS).

Don't circle the x's that don't help.

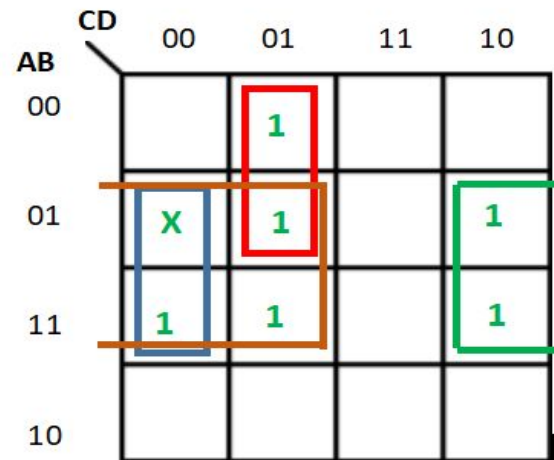
BC					
A		00	01	11	10
0	0	X	1	1	2
1	4		X	1	6

Reduced form : $F = C$

DON'T CARE IN KARNAUGH MAP

1. Minimise the following function in SOP minimal form using K-Maps: $f(A,B,C,D) = m(1, 5, 6, 12, 13, 14) + d(4)$

- From **green & blue** group we find terms : BD'
- From **red** group we find terms : $A'C'D$
- From **brown** group we find terms : BC'
- Therefore, SOP minimal is, $f = BC' + BD' + A'C'D$



DON'T CARE IN KARNAUGH MAP

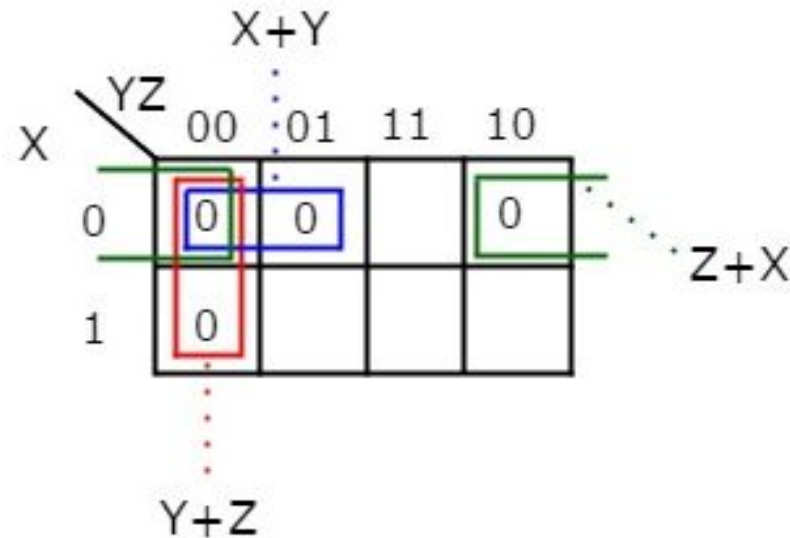
2. Minimise the following function in SOP minimal form using K-Maps: $F(A, B, C, D) = m(1, 2, 6, 7, 8, 13, 14, 15) + d(3, 5, 12)$

- From **green** group we find terms : AB
- From **red** group we find terms : $AC'D'$
- From **brown** group we find terms : $A'D$
- From **blue** group we find terms : $A'C$
- $f = AC'D' + A'D + A'C + AB$

AB \ CD	CD			
	00	01	11	10
00		1	X	1
01		X	1	1
11	X	1	1	1
10	1			

KARNAUGH MAP: POS

Eg: $F(X,Y,Z) = \prod M(0,1,2,4)$ minimize the given using K-MAP in POS form.



$$\text{Ans: } F' = X'Z' + Y'Z' + X'Y'$$

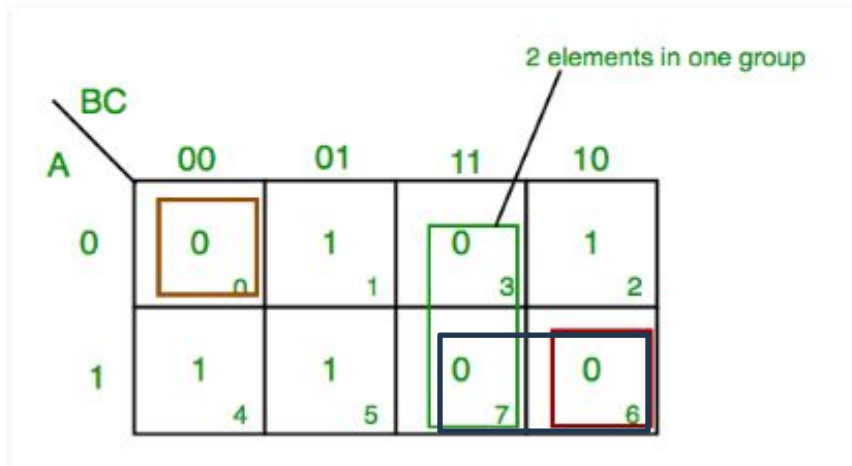
$$(F')' = F = (X'Z' + Y'Z' + X'Y')'$$

$$F = (X + Z) \cdot (Y + Z) \cdot (X + Y)$$

KARNAUGH MAP: POS

1. K-map of 3 variables-

$$F(A,B,C)=\pi(0,3,6,7)$$



- $F' = BC + A'B'C' + AB$
 $(F')' = F = (B.C + A'B'C' + ABC')' = (B' + C')(A + B + C)(A' + B')$

KARNAUGH MAP: POS

Eg: Minimise the following function in POS minimal form using K-Maps: $F(A, B, C, D) = M(6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$

- For **red** group we find terms: $(B' + C')$
- For **green** group we find terms: A'
- POS minimal is, $F' = A + BC$

$$\begin{aligned}(F')' &= (A + BC)' \\ &= A'(B' + C')\end{aligned}$$

		CD			
		00	01	11	10
AB	00				
	01			0	0
	11	X	X	X	X
	10	0	0	X	X

DON'T CARE IN KARNAUGH MAP

Eg: $F(A,B,C,D) = \sum m(4,5,7,8,10,11,13,14) + \sum d(0,1,2)$ minimize the given using K-MAP in POS form.

CD \ AB	00	01	11	10
00	X ⁰	X ¹	0 ³	X ²
01	1 ⁴	1 ⁵	1 ⁷	0 ⁶
11	0 ¹²	1 ¹³	0 ¹⁵	1 ¹⁴
10	1 ⁸	0 ⁹	1 ¹¹	1 ¹⁰

Ans: $F' = ABCD + ABC'D' + A'CD' + B'C'D + A'B'$
 $F = (A+B) (B+C+\bar{D}) (A+\bar{C}+D) (\bar{A}+\bar{B}+C+D) (\bar{A}+\bar{B}+\bar{C}+D)$

FIVE-VARIABLE KARNAUGH MAP

- Maps for more than four variables are not as simple to use. A five-variable map needs 32 squares and a six-variable map needs 64 squares.
- When the number of variables becomes large, the number of squares becomes excessively large and the geometry for combining adjacent squares becomes more involved.
- The five-variable map is shown in Figure. It consists of 2 four-variable maps with variables A, B, C, D, and E.
- Variable A distinguishes between the two maps, as indicated on the top of the diagram.
- The left-hand four-variable map represents the 16 squares where $A = 0$, and the other four-variable map represents the squares where $A = 1$. Minterms 0 through 15 belong with $A = 0$ and minterms 16 through 31 with $A = 1$.

FIVE-VARIABLE KARNAUGH MAP

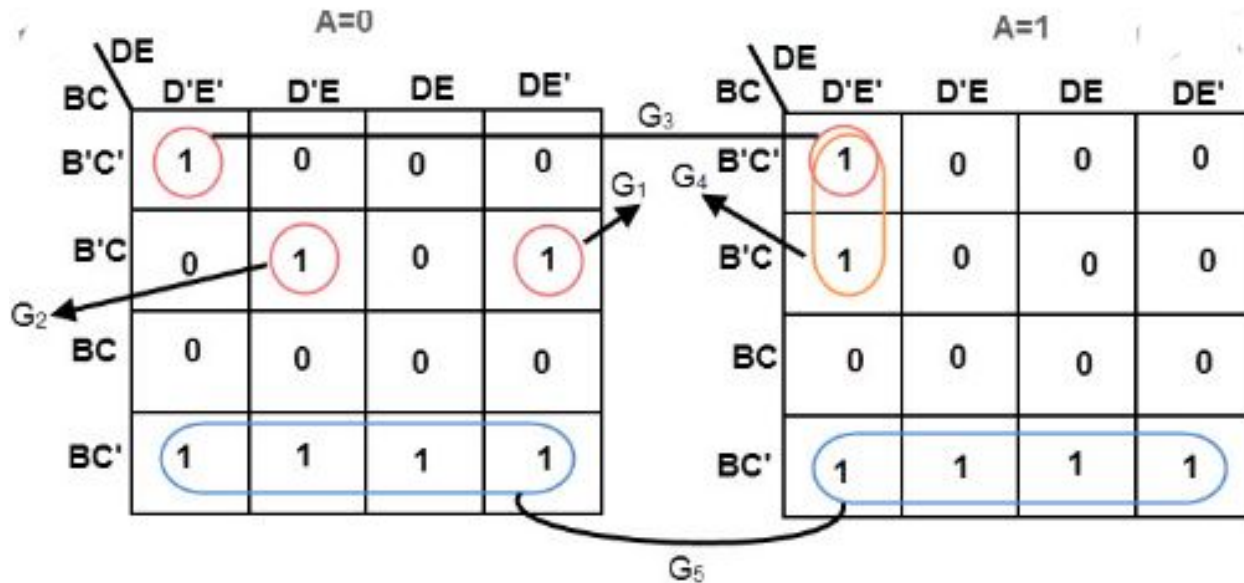
A	B	C	D	E	minterm	Notation
0	0	0	0	0	$A'B'C'D'E'$	m_0
0	0	0	0	1	$A'B'C'D'E$	m_1
...	...	...	...	...		
1	1	1	1	0	$ABCDE'$	m_{30}
1	1	1	1	1	$ABCDE$	m_{31}

		$\bar{A}$						A			
		DE						DE			
		00	01	11	10			00	01	11	10
BC	00	0	1	3	2	BC		16	17	19	18
	01	4	5	7	6			20	21	23	22
	11	12	13	15	14			28	29	31	30
	10	8	9	11	10			24	25	27	26

Kmap for 5 variables

FIVE-VARIABLE KARNAUGH MAP

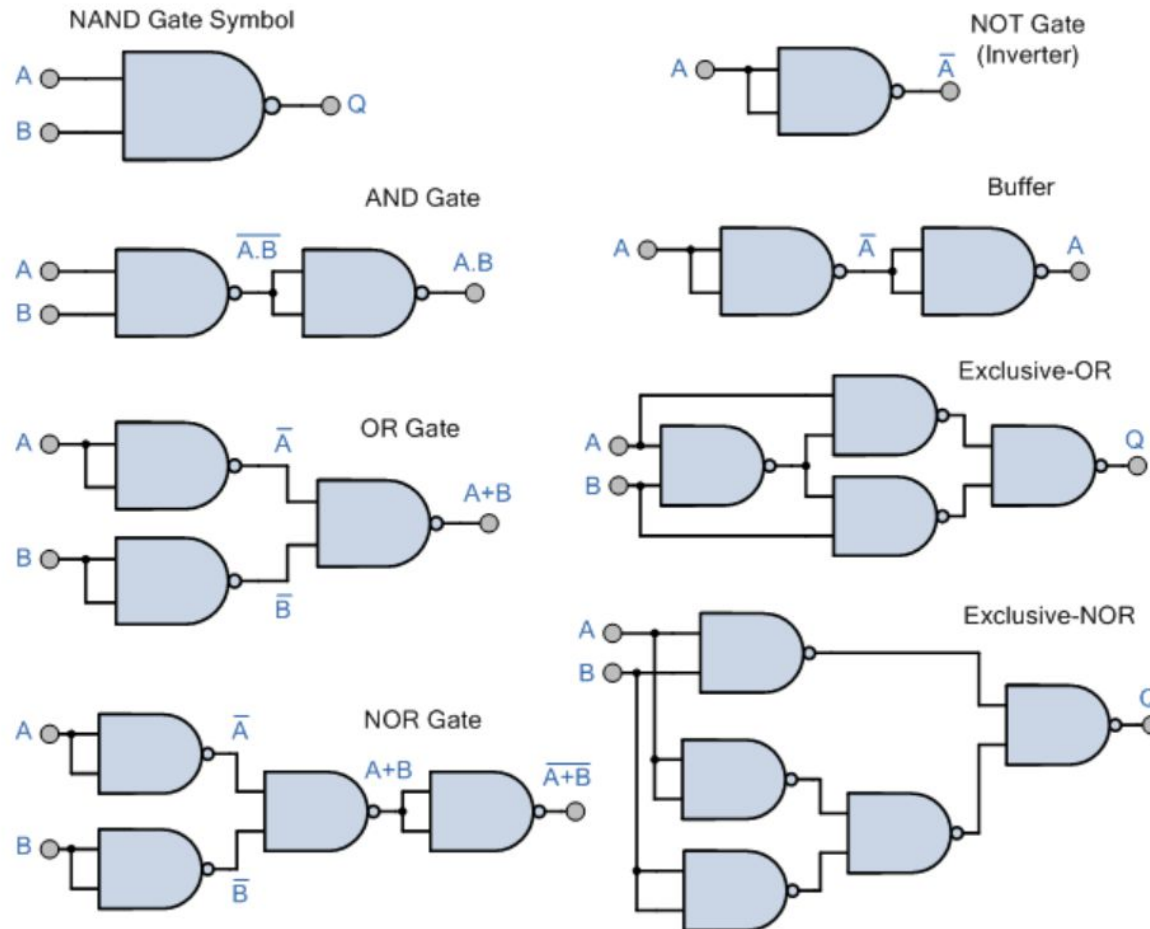
Eg: $F(A,B,C,D,E) = \sum m(0, 5, 6, 8, 9, 10, 11, 16, 20, 24, 25, 26, 27)$
minimize the given using K-MAP in SOP form.



Sol: $F = A'B'CDE' + A'B'CD'E + B'C'D'E' + AB'D'E' + BC'$

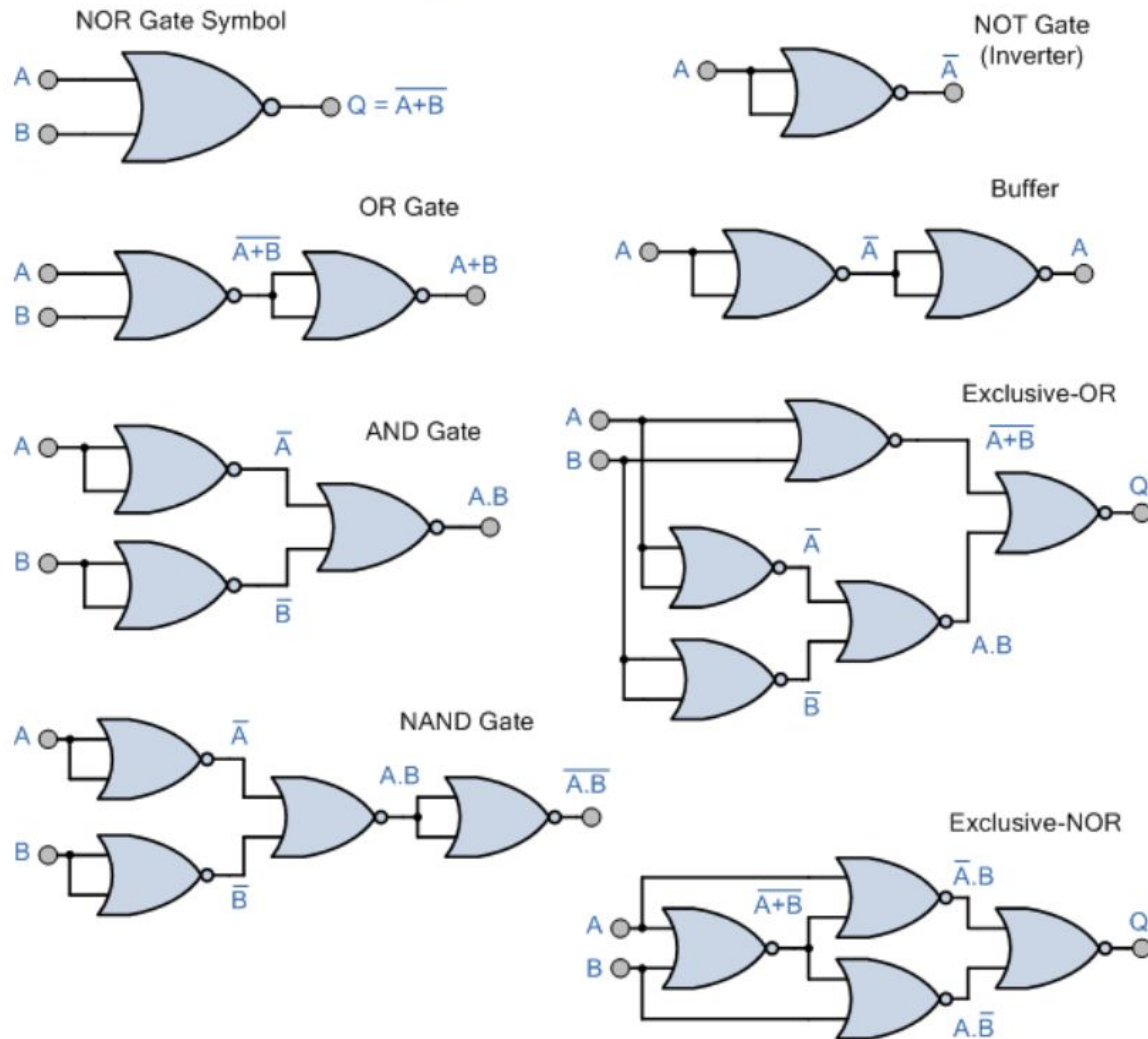
Realization of Logic Gate Using Universal gates

Universal Logic Gates using only NAND Gates



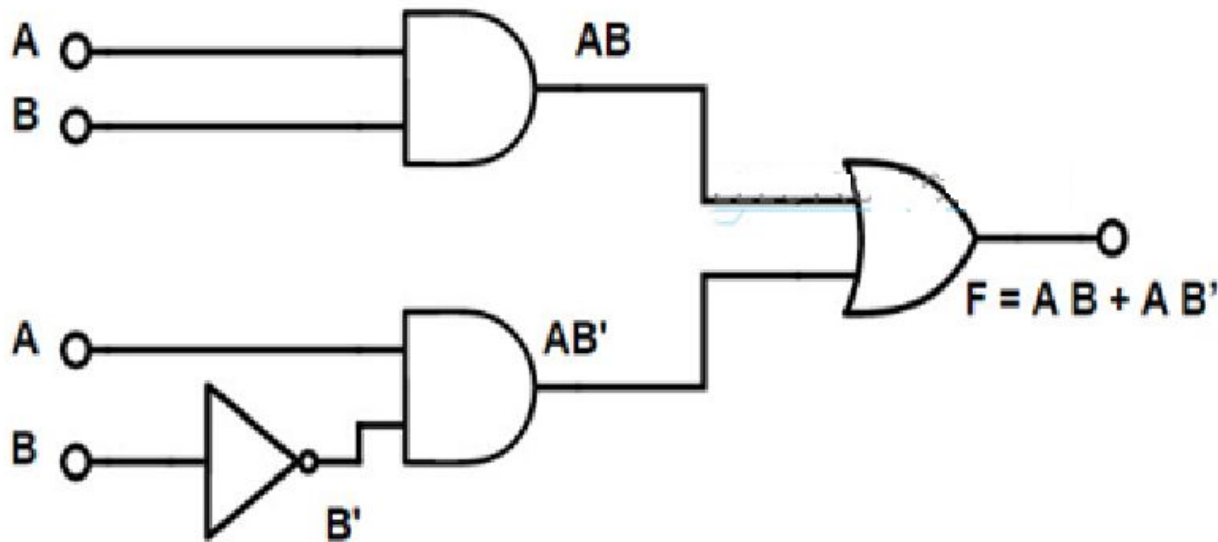
Realization of Logic Gate Using Universal gates

Universal Logic Gates using only NOR Gates



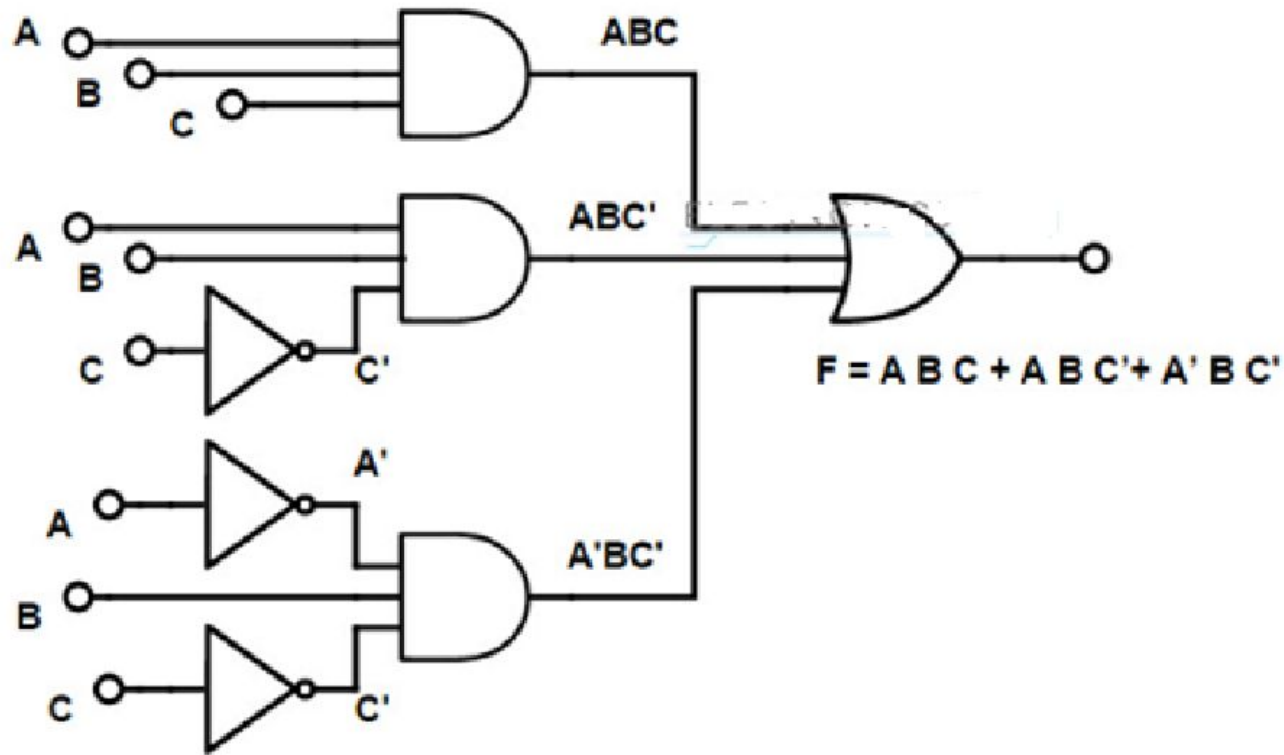
Implementation of 2 Input Variables Boolean Function using Logic Gates

- $F = AB + AB'$



Implementation of 3 Input Variables Boolean Function using Logic Gates

- $F = A B C + A B C' + A' B C'$



Daily Quiz

- Complement of the function $F = x'y'z + xy'z'$ is:
 - a) $(x + y + z'). (x' + y + z)$
 - b) $(x' + y' + z). (x + y' + z')$
 - c) $(x + y + z). (x' + y' + z)$
- Minimized function for $F(A, B, C) = \Sigma (2,3,6) + d(0,7)$ using k-map is: (a) $A'C' + BC + BC'$ (b) B (c) AB

Weekly Assignment

- $F(A,B,C) = \sum m(1,4,5,7)$ minimize the given using K-MAP in SOP form.
- $F(A,B,C,D) = \sum m(0,2,4,6,8,10,12,14)$ minimize the given using K-MAP in SOP form.
- $F(w,x,y,z) = \sum m(4,5,7,8,10,14)$ minimize the given using K-MAP in POS form.
- $F(A,B,C,D,E) = \sum m(4,5,7,8,10,14,27,31) + \sum d(0,1,2,11,19,25)$ minimize the given using K-MAP in SOP form.
- $F(w,x,y,z) = \sum m(0,2,4,5,8,12,14) + \sum d(1,3,5,7,9)$ minimize the given using K-MAP in POS form.

Weekly Assignment

- Convert the following numbers
 - i) $(163.789)_{10}$ to Octal number
 - ii) $(11001101.0101)_2$ to base-8 and base-4
 - iii) $(4567)_{10}$ to base 2
 - iv) $(4D.56)_{16}$ to Binary
- Subtract $(111001)_2$ from $(101011)_2$ using 1's complement?
- perform $(-50)-(-10)$ in binary using the signed-2's complement
- Determine the value of base x if $(211)_x = (152)_8$
- Convert the following numbers
 - i) $(250.5)_{10} = ()_2$
 - ii) $(673.23)_{10} = ()_8$
 - iii) $(101110.01)_2 = ()_8$
- Convert the following to binary and then to gray code $(AB33)_{16}$
- Perform the following Using BCD arithmetic $(7129)_{10} + (7711)_{10}$
- What is Digital System? Characteristics of digital systems. Explain the difference between analog and digital systems.

Weekly Assignment

- Design the circuit by Using NAND gates $F = ABC' + DE + AB'D'$
- Simplify and implementation the following SOP function using NOR gates $F(A,B,C,D) = \sum m(0,1,4,5,10,11,14,15)$
- Convert the following
 - a) $(1AD)_{16} = ()_{10}$
 - b) $(453)_8 = ()_{10}$
 - c) $(10110011)_2 = ()_{10}$
 - d) $(5436)_{10} = ()_3$
- State and Explain the DeMorgan's Theorem and Consensus Theorem
- Convert the following numbers
 - a) $(615)_{10} = ()_{16}$
 - b) $(214)_{10} = ()_8$
 - c) $(0.8125)_{10} = ()_2$
 - d) $(658.825)_{10} = ()_8$ v) $(54)_{10} = ()_2$

MCQs

1. The number system which uses alphabets as well as numerals is Binary number system ?
 - Octal number system
 - Decimal number system
 - Hexadecimal number system

2. The number that come immediately after and before hex number (FFEF)₁₆ are respectively
 - (FFF0)₁₆ and (FFE0)₁₆
 - (FFFE)₁₆ and (FFEE)₁₆
 - (FFF0)₁₆ and (FFFE)₁₆
 - (FFF0)₁₆ and (FFEE)₁₆

MCQs

- 4 Two input exclusive NOR gate gives high output
- When one input is high and the output is low
 - Only when both the inputs are low
 - When both the inputs are same
 - Only when both the inputs are high
5. What is 2's complement or 0011 0101 1001 1100 number?
- 1100 1010 1100 1011
 - 1100 1010 0110 0011
 - 1100 1010 0110 0100
 - None of these

MCQs

5. The maxterm designator of the term $A' + B' + C + D'$ is

- 2
- 13
- 10
- None of these

6. The minterm, designator of the term $AB'CD$ is

- 4
- 15
- 11
- None of these

Glossary questions

- The minterm corresponding to decimal number 15 is -----
- Decimal digit 5 is represented by -----using 7312 weighted code.
- Represent the decimal number $(0.875)_{10}$ into binary form.
- A decimal number 6 in excess-3 code is-----
- $(1111)_2 - (1111)_2$ is equal to-----
- 9's complement of 5436 is-----
- $(28CE5)_H + (AB2C3)_H = (\text{-----})_H$
- Characteristic of Gray code is -----,-----and it is unit -----code.
- For the identity $AB + A'C + BC = AB + A'C$, the dual form is-----
- The minimum number of NAND gates required to implement $A + AB' + AB'C$ is equal to-----

Cont...

- The code used for labelling the cells of the K-map is-----
- Switches connected in parallel behave as-----
- Output of gate -----is $(A'B)'$
- Positive logic in a logic circuit is one in which logic 0 voltage level is-----than logic 1 voltage level.
- AND-OR realization is equivalent to -----realization.
- 2-input EX-OR gate can also be performed as inverter if its one of the input is fixed as logic-----
- The output of a logic gate is '1' when all its input are at logic '0' the gate is either----- or an -----
- Given the logic function of Four variables $f(A,B,C,D) = (A' + BC)(B + CD)$. The function as a sum of product will be-----

Old Questions

- Minimize the following function by Quine McClusky method and also perform the NAND implementation of the simplified function.

$$F(w,x,y,z) = \sum m(1,4,8,9,13,14,15) + d(2,3,11,12)$$

- $F(A,B,C,D) = \sum m(0,1,2,3,10,11,12,13,14,15)$ minimize the given using QM method.
- $F(A,B,C,D) = \sum m(1,3,7,11,15) + \sum d(0,2,5)$ minimize the given using QM method.

Minterm and Maxterms

- In a similar fashion, n variables forming an OR term, with each variable being primed or unprimed, provide 2^n possible combinations, called *maxterms*, or *standard sums*.
- The eight maxterms for three variables, together with their symbolic designations, are listed in Table 2.3 . Any 2^n maxterms for n variables may be determined similarly. It is important to note that
 - each maxterm is obtained from an OR term of the n variables, with each variable being unprimed if the corresponding bit is a 0 and primed if a 1.
 - each maxterm is the complement of its corresponding minterm and vice versa.

Table 2.3
Minterms and Maxterms for Three Binary Variables

x	y	z	Minterms		Maxterms	
			Term	Designation	Term	Designation
0	0	0	$x'y'z'$	m_0	$x + y + z$	M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

SOP POS forms

- A Boolean function can be expressed algebraically from a given truth table by forming a minterm for each combination of the variables that produces a 1 in the function and then taking the OR of all those terms.
- For example, the function $f_1=1$ in Table 2.4 is determined by expressing the combinations 001, 100, and 111 as $x'y'z$, $xy'z'$, and xyz , respectively. Since each one of these minterms results in $f_1 = 1$, we have

$$f_1 = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7$$

Table 2.4
Functions of Three Variables

<i>x</i>	<i>y</i>	<i>z</i>	Function f_1	Function f_2
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

SOP POS contd...

Similarly, it may be easily verified that

$$f_2 = x'yz + xy'z + xyz' + xyz = m_3 + m_5 + m_6 + m_7$$

These examples demonstrate an important property of Boolean algebra: Any Boolean function can be expressed as a sum of minterms (with “sum” meaning the ORing of terms).

Now consider the complement of a Boolean function. It may be read from the truth table by forming a minterm for each combination that produces a 0 in the function and then ORing those terms. The complement of f_1 is read as

$$f_1' = x'y'z' + x'yz' + x'yz + xy'z + xyz'$$

If we take the complement of f_1' , we obtain the function f_1 :

$$\begin{aligned} f_1 &= (x + y + z)(x + y' + z)(x' + y + z')(x' + y' + z) \\ &= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6 \end{aligned}$$

Similarly, it is possible to read the expression for f_2 from the table:

$$\begin{aligned} f_2 &= (x + y + z)(x + y + z')(x + y' + z)(x' + y + z) \\ &= M_0 M_1 M_2 M_4 \end{aligned}$$

These examples demonstrate a second property of Boolean algebra: Any Boolean function can be expressed as a product of maxterms (with “product” meaning the ANDing of terms). The procedure for obtaining the product of maxterms directly from the truth table is as follows: Form a maxterm for each combination of the variables that produces a 0 in the function, and then form the AND of all those maxterms. **Boolean functions expressed as a sum of minterms or product of maxterms are said to be in *canonical form*.**

Sum of Minterms

Previously, we stated that, for n binary variables, one can obtain 2^n distinct minterms and that any Boolean function can be expressed as a sum of minterms. **The minterms whose sum defines the Boolean function are those which give the 1's of the function in a truth table.** Since the function can be either 1 or 0 for each minterm, and since there are 2^n minterms, one can calculate all the functions that can be formed with n variables to be 2^{2^n} . It is sometimes convenient to express a Boolean function in its sum-of-minterms form. If the function is not in this form, it can be made so by first expanding the expression into a sum of AND terms. Each term is then inspected to see if it contains all the variables. If it misses one or more variables, it is ANDed with an expression such as $x + x'$, where x is one of the missing variables. The next example clarifies this procedure.

Product of Maxterms

Each of the 2^{2^n} functions of n binary variables can be also expressed as a product of maxterms. To express a Boolean function as a product of maxterms, it must first be brought into a form of OR terms. This may be done by using the distributive law, $x + yz = (x + y)(x + z)$. Then any missing variable x in each OR term is ORed with xx' . The procedure is clarified in the following example.

EXAMPLE 2.4

Express the Boolean function $F = A + B'C$ as a sum of minterms. The function has three variables: A , B , and C . The first term A is missing two variables; therefore,

$$A = A(B + B') = AB + AB'$$

This function is still missing one variable, so

$$\begin{aligned} A &= AB(C + C') + AB'(C + C') \\ &= ABC + ABC' + AB'C + AB'C' \end{aligned}$$

The second term $B'C$ is missing one variable; hence,

$$B'C = B'C(A + A') = AB'C + A'B'C$$

Combining all terms, we have

$$\begin{aligned} F &= A + B'C \\ &= ABC + ABC' + AB'C + AB'C' + A'B'C \end{aligned}$$

But $AB'C$ appears twice, and according to theorem 1 ($x + x = x$), it is possible to remove one of those occurrences. Rearranging the minterms in ascending order, we finally obtain

$$\begin{aligned} F &= A'B'C + AB'C + AB'C' + ABC' + ABC \\ &= m_1 + m_4 + m_5 + m_6 + m_7 \end{aligned}$$

When a Boolean function is in its sum-of-minterms form, it is sometimes convenient to express the function in the following brief notation:

$$F(A, B, C) = \Sigma(1, 4, 5, 6, 7)$$

The summation symbol Σ stands for the ORing of terms; the numbers following it are the indices of the minterms of the function. The letters in parentheses following F form a list of the variables in the order taken when the minterm is converted to an AND term.

An alternative procedure for deriving the minterms of a Boolean function is to obtain the truth table of the function directly from the algebraic expression and then read the minterms from the truth table. Consider the Boolean function given in Example 2.4:

$$F = A + B'C$$

The truth table shown in Table 2.5 can be derived directly from the algebraic expression by listing the eight binary combinations under variables A , B , and C and inserting 1's under F for those combinations for which $A = 1$ and $BC = 01$. From the truth table, we can then read the five minterms of the function to be 1, 4, 5, 6, and 7.

Table 2.5
Truth Table for $F = A + B'C$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

EXAMPLE 2.5

Express the Boolean function $F = xy + x'z$ as a product of maxterms. First, convert the function into OR terms by using the distributive law:

$$\begin{aligned} F &= xy + x'z = (xy + x')(xy + z) \\ &= (x + x')(y + x')(x + z)(y + z) \\ &= (x' + y)(x + z)(y + z) \end{aligned}$$

The function has three variables: x , y , and z . Each OR term is missing one variable; therefore,

$$\begin{aligned} x' + y &= x' + y + zz' = (x' + y + z)(x' + y + z') \\ x + z &= x + z + yy' = (x + y + z)(x + y' + z) \\ y + z &= y + z + xx' = (x + y + z)(x' + y + z) \end{aligned}$$

Combining all the terms and removing those which appear more than once, we finally obtain

$$\begin{aligned} F &= (x + y + z)(x + y' + z)(x' + y + z)(x' + y + z') \\ &= M_0 M_2 M_4 M_5 \end{aligned}$$

A convenient way to express this function is as follows:

$$F(x, y, z) = \Pi(0, 2, 4, 5)$$

The product symbol, Π , denotes the ANDing of maxterms; the numbers are the indices of the maxterms of the function.

Conversion between Canonical Forms

The complement of a function expressed as the sum of minterms equals the sum of minterms missing from the original function. This is because the original function is expressed by those minterms which make the function equal to 1, whereas its complement is a 1 for those minterms for which the function is a 0. As an example, consider the function

$$F(A, B, C) = \Sigma(1, 4, 5, 6, 7)$$

This function has a complement that can be expressed as

$$F'(A, B, C) = \Sigma(0, 2, 3) = m_0 + m_2 + m_3$$

Now, if we take the complement of F' by DeMorgan's theorem, we obtain F in a different form:

$$F = (m_0 + m_2 + m_3)' = m_0' \cdot m_2' \cdot m_3' = M_0 M_2 M_3 = \Pi(0, 2, 3)$$

The last conversion follows from the definition of minterms and maxterms as shown in Table 2.3. From the table, it is clear that the following relation holds:

$$m_j' = M_j$$

That is, the **maxterm with subscript j is a complement of the minterm with the same subscript j and vice versa.**

The last example demonstrates the conversion between a function expressed in sum-of-minterms form and its equivalent in product-of-maxterms form. A similar argument will show that the conversion between the product of maxterms and the sum of minterms is similar. We now state a general conversion procedure: To convert from one canonical form to another, interchange the symbols Σ and Π and list those numbers missing from the original form. In order to find the missing terms, one must realize that the total number of minterms or maxterms is 2^n , where n is the number of binary variables in the function.

A Boolean function can be converted from an algebraic expression to a product of maxterms by means of a truth table and the canonical conversion procedure. Consider, for example, the Boolean expression

$$F = xy + x'z$$

First, we derive the truth table of the function, as shown in Table 2.6. The 1's under F in the table are determined from the combination of the variables for which $xy = 11$ or $xz = 01$. The minterms of the function are read from the truth table to be 1, 3, 6, and 7. The function expressed as a sum of minterms is

$$F(x, y, z) = \Sigma(1, 3, 6, 7)$$

Since there is a total of eight minterms or maxterms in a function of three variables, we determine the missing terms to be 0, 2, 4, and 5. The function expressed as a product of maxterms is

$$F(x, y, z) = \Pi(0, 2, 4, 5)$$

Table 2.6

Truth Table for $F = xy + x'z$

x	y	z	F	
0	0	0	0	
0	0	1	1	
0	1	0	0	
0	1	1	1	
1	0	0	0	
1	0	1	0	
1	1	0	1	
1	1	1	1	
				Maxterms